JIAQI (JACKY) YANG

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SUMMARY

PhD researcher (The Hong Kong Polytechnic University & University of Surrey) who combines large language models, machine learning and causal inference to measure behaviour from text and to estimate what actually drives outcomes. Builds LLM pipelines that run daily, check their own sources and are measured against test sets. Two Best Paper awards in 2026; papers under review at the leading journals in tourism research.

EDUCATION

The Hong Kong Polytechnic University & University of Surrey — Hong Kong SAR & Guildford, UK
Dual PhD, Hotel and Tourism Management / Surrey Business School — *Sept. 2024 – Expected July 2027*

Shanghai University of International Business and Economics — Shanghai, China
M.Sc. in Applied Statistics — GPA 3.81/4.0, rank 5/111 — *2022 – 2024*

Shanghai University & University of Technology Sydney (joint programme) — Shanghai, China
B.Econ. in Finance (GPA 3.58/4.0) and B.Bus. (GPA 6.44/7.0) — *2017 – 2021*

SKILLS

LLMs & AI	Gemini, OpenAI and Claude APIs, structured output, retrieval-augmented generation (BM25, vector, hybrid), tool-using agents, LLM annotation and its validation (human agreement, Krippendorff's α), multimodal (video, image) extraction
AI workflow	Claude Code for research and coding: jackyyangjq.github.io/ai-workflow
Data science	Causal inference (staggered difference-in-differences, event study, synthetic control), time-series and panel econometrics, quantile regression, forecasting, machine learning (XGBoost, random forest, LASSO, SHAP), NLP and sentiment analysis
Engineering	Python, R, Stata, SQL (SQLite FTS5), Git, GitHub Actions CI, automated testing, LaTeX
Languages	Mandarin Chinese (native); English (IELTS 7.5)

SELECTED PROJECTS

llm-text-measurement — validated LLM text measurement — GitHub
Python, GPT-5.5, Claude, scikit-learn, Word2Vec — *2026*

- Labelled 998 public hotel reviews with aspects, polarity and a verbatim quote for every label; labels whose quote is not in the text are rejected, and doubtful answers get a second pass with retrieved examples.
- Validated six ways: 94.6% agreement with guests' own liked/disliked split (Cohen's κ 0.89), Krippendorff's α 0.96 on a repeat run and 0.93 against a second model, no false labels on 60 controls.

filings-qa-agent — question answering over SEC filings — GitHub
Python, Gemini, SQLite FTS5, numpy — *2026*

- Indexed 48 annual and quarterly reports of 12 large US companies; every answer sentence cites a source passage, and sentences citing passages the model was not shown are dropped automatically.
- Evaluated keyword, vector and hybrid retrieval on 50 questions: BM25 95% correct, hybrid 90%, vector 70%; the system declined all 30 unanswerable question pairs.
- Added a traced research agent that chooses between filings, price and news tools; 168 offline tests, CI and a key-free offline demo.

ml-inflation-forecasting — machine-learning inflation forecasts — GitHub
Python, scikit-learn, XGBoost, SHAP — *2026*

- Replicated and extended Medeiros et al. (2021, *JBES*): 8 models refitted monthly on 121 FRED-MD series over 308 rolling windows, with Diebold–Mariano tests, model confidence sets and SHAP explanations.
- Tree models cut 12-month forecast error by 15–20% against an AR benchmark in 2001–2015, but no model beat it after 2020.

finfluencer-digest — multimodal LLM research digest GitHub
Python, Gemini, GitHub Actions *2026*

- Daily pipeline in which Gemini watches videos from 16 YouTube finance channels and reads X posts, extracts structured stock calls, aggregates consensus and disagreement, and emails a digest.
- Multi-model fallback, a guard against dropped scheduled runs, and source-verified news links to prevent fabricated citations; 27 offline tests, CI and a demo mode.

applied-stats-econometrics-toolkit — statistics and econometrics in Python GitHub
Python, statsmodels, pyfixest *2026*

- Staggered DiD (Callaway–Sant'Anna, Sun–Abraham, Goodman-Bacon), fixed-effects quantile regression, VAR, Tobit, Oster bounds and change points, each checked against simulated truth.

Open-source contributions GitHub
Python (FastAPI), TypeScript *2026*

- Extended a local-first portfolio dashboard with broker account sync and a workspace that scores stock calls against later prices; four pull requests opened upstream.
- Merged fix to the ClaudeCodeUsage VS Code extension. Also built a vision-model calorie tracker web app (20 versions in seven weeks, calibrated against real weight data).

RESEARCH EXPERIENCE

The Hong Kong Polytechnic University & University of Surrey Sept. 2024 – present
Doctoral researcher, AI and data in tourism and hospitality *Hong Kong SAR & Guildford, UK*

- **LLM text measurement at scale:** co-developed LLM-based aspect-level sentiment measures from online reviews and tested how well they predict tourist demand (under review, *Journal of Travel Research*).
- **LLM classification of corporate disclosures:** separated automation- from augmentation-oriented digital strategies in firm texts and estimated their different effects on performance (CHME 2026 Overall Best Paper Award).
- **Causal policy evaluation:** used the staggered rollout of visa and tourism-promotion policies as natural experiments to estimate effects on local housing markets (major revision, *Annals of Tourism Research*, corresponding author; IATE 2026 Best Paper Award).
- **NLP index construction:** built a firm-level green transformation index from annual-report text and evaluated China's carbon trading pilots with difference-in-differences (*Journal of Cleaner Production*, 2026, first author).

AWARDS

- Best Paper Award, International Association for Tourism Economics Conference (IATE 2026) 2026
- Overall Best Paper Award and Hospitality Management Track Best Paper, CHME 2026 2026
- National First Prize, National College Student Market Research and Analysis Competition 2020
- National Second Prize, National College Student Statistical Modeling Competition 2023
- Outstanding Graduate of Shanghai Municipality 2024

TEACHING & LEADERSHIP

- Teaching Assistant, Financial Management in Hospitality (PolyU): led three weekly tutorial sections, designed spreadsheet exercises, assessed group projects. 2025
- Guest lecturer, Surrey Business School: AI for study and research (MSc Research Methods, three sessions), AI in digital marketing (summer camp) and smart check-in technology (MSc). 2026
- Moderator, panel "From Graduate to Scholar", 21st Asia Pacific Forum for Graduate Students Research in Tourism. 2025